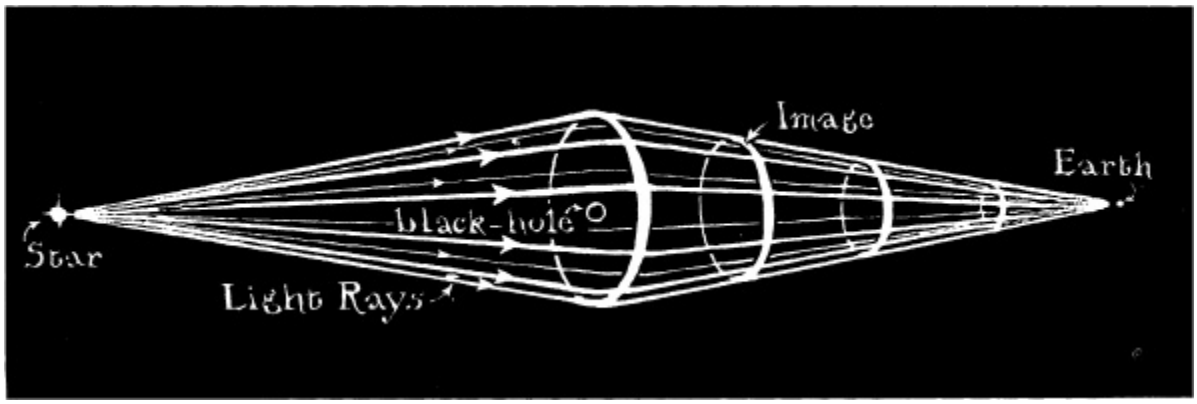
How could an astronomer possibly detect a black hole at such a great distance? Could an astronomer just watch the sky for a moving, dark object which blots out the light from stars behind it? No. With its circumference of roughly 50 kilometers and its distance of at least 4 light-years, the hole's dark disk will subtend an angle no larger than 10^{-7} arc second. That is roughly the thickness of a human hair as seen from the distance of the Moon, and 10 million times smaller than the resolution of the world's best telescopes. The moving dark object would be invisibly tiny.

If one could not see the hole's dark disk as the hole goes in front of a star, might one see the hole's gravity act like a lens to magnify the star's light (Figure 8.2)? Might the star appear dim at first, then brighten as the hole moves between Earth and the star, then dim again as the hole moves on? No, this method of search also will fail. The reason it will fail depends on whether the star and the hole are orbiting around each other and thus are close together, or are separated by typical interstellar distances. If they are close together, then the tiny hole will be like a hand-held magnifying glass placed upright on a windowsill on the eighty-ninth floor of the Empire State Building, and then viewed from several kilometers distance. Of course, the tiny magnifying glass has no power to magnify the building's appearance, and similarly the hole has no effect on the star's appearance.



8.2 A black hole's gravity should act like a lens to change the apparent size and shape of a star as seen from Earth. In this figure the hole is precisely on the line between the star and the Earth, so light rays from the star can reach the Earth equally well by going over the top of the hole, or under the bottom, or around the front, or around the back. All the light rays reaching Earth move outward from the star on a diverging cone; as they pass the hole they get bent down toward Earth; they then arrive at Earth on a converging cone. The resulting image of the star on the Earth's sky is a thin ring. This ring has far larger surface area, and hence far larger total brightness, than the star's image would have if the black hole were absent. The ring is too